

Radar Signal Classification Using Machine Learning and Neural Networks

Student: Sitah Al-Yahya | Supervisor: Raghad Al-Shammari

Abstract

This project explores the application of machine learning and neural networks to classify radar signals. By analyzing signal features, we aim to improve accuracy and efficiency in signal identification compared to traditional methods.

Introduction

Radar signal classification is critical for defense and civilian applications. Traditional methods often struggle with complex signal environments. Our approach leverages deep learning to handle non-linear patterns effectively.

Goal

To develop a robust classification model capable of distinguishing between various radar signal types with high accuracy using deep learning techniques.

Dataset Description

The dataset consists of various radar pulse characteristics, normalized for input into the neural network architecture.

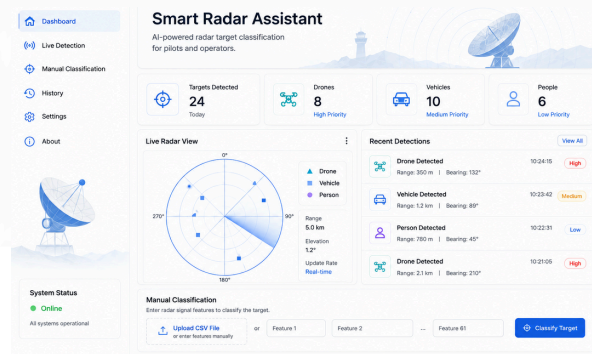
Methodology

- Data Preprocessing.
- Model Selection (LR, RF, KNN, DT, NN).
- Training and Validation.
- Performance Evaluation.

Conclusion

The Neural Network model outperformed traditional machine learning algorithms, demonstrating superior capabilities for radar signal classification.

System Design



Implementation Tools

Python, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras

Results & Comparison

Model	Accuracy
Logistic Regression	0.570
Random Forest	0.748
KNN	0.759
Decision Tree	0.747
Neural Network	0.789

